Jozef Sawan

Master (Bac+5) — Mechanical and Materials Engineering

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Mechanical and Materials engineer passionate about design and simulation. I thrive in environments where research, creativity, and technical rigor come together to deliver innovative solutions.

Education

2018–2023 **Université Grenoble Alpes**, *BSc in Mechanical Engineering and Industrial Engineering (GMP)*

2023–2025 **Université Bretagne Sud**, *MEng / Master's in Mechanical and Materials Engineering (GMM)*

Experience

Dec 2022 – Sep 2023 **Software Engineering Intern**, *iGoFlex*
Contributed to the development of an embedded application. Developed the e-commerce website. Automated order integration into a database. Redesigned the company website.

Jul – Aug 2022 **Software Engineering Intern**, *Phonexpat*
Developed an automated configuration tool for customer modems. Provided English-language support for installation and configuration of connected devices (IoT). Redesigned the company website.

Apr – Jun 2022 **Quality Intern**, *Artxbat*
Rewrote and updated the Quality Assurance Manual. Created tracking sheets for operations.

Mar – Jul 2024 **Startup Creation — Pépite Program**, *JoCard*
Founded a company in Lorient under the French National Student-Entrepreneur Status (SNEE).

Feb – Aug 2025 **R&D Intern — Vibration and Acoustics**, *VENATHEC*
Finite-element modeling in COMSOL of structures and vibration propagation studies.
Modal analysis, admittance, correlation/validation between simulations and measurements, developed Python scripts for post-processing.
Optimized multi-source vibration simulation methodologies.
Participated in acoustic and vibration measurement campaigns.

Skills

Strength of Materials
Bending, tension, torsion, shear, cut method and integration techniques, stress and strain in structures.

Structural Dynamics
Modal analysis, free and forced vibrations, transient response, dynamic behavior of plates and beams.

Finite Element Method (FEM)
1D/2D/3D modeling, appropriate meshing, boundary conditions, computation time optimization, results validation.

Software

SolidWorks, Abaqus, COMSOL Multiphysics, ANSYS Workbench, CES EduPack, ESPRIT TNG
Python, MATLAB, RDM6, LaTeX, Microsoft Office, Google tools (Ads, Analytics, GTM...), WordPress
Photoshop, Blender, graphic and web creation tools, entrepreneurship skills

Languages

French — Native | English — C2 (Fluent) | German — A2 (Goethe-Institut)
Arabic (Lebanese) — Basic knowledge

Interests

Sports: Volleyball, tennis, badminton, archery, hiking, fishing
Making/Creation: 3D printing, graphic design, 3D modeling (Blender), website creation
Travel: Germany, Austria, Portugal, Spain, United States, Italy, Lebanon, United Kingdom, Turkey